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# Risk management—
# Risk assessment techniques

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## Foreword

This translation has been made based on the original Japanese Industrial Standard established by the Minister of Economy, Trade and Industry through deliberations at the Japanese Industrial Standards Committee according to the proposal for establishment of Japanese Industrial Standard submitted by Japanese Standards Association (JSA) with the draft being attached, based on the provision of Article 12 Clause 1 of the Industrial Standardization Law.

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# Risk management—Risk assessment techniques

## Introduction

This Japanese Industrial Standard has been prepared based on the first edition of **IEC/ISO 31010** published in 2009 without any modifications of technical contents.

Organizations of all types and sizes face a range of risks that may affect the achievement of their objectives.

These objectives may relate to a range of the organization's activities, from strategic initiatives to its operations, processes and projects, and be reflected in terms of societal, environmental, technological, safety and security outcomes, commercial, financial and economic measures, as well as social, cultural, political and reputation impacts.

All activities of an organization involve risks that should be managed. The risk management process aids decision making by taking account of uncertainty and the possibility of future events or circumstances (intended or unintended) and their effects on agreed objectives.

Risk management includes the application of logical and systematic methods for

— communicating and consulting throughout this process;

— establishing the context for identifying, analysing, evaluating, treating risk associated with any activity, process, function or product;

— monitoring and reviewing risks;

— reporting and recording the results appropriately.

Risk assessment is that part of risk management which provides a structured process that identifies how objectives may be affected, and analyses the risk in terms of consequences and their probabilities before deciding on whether further treatment is required.

Risk assessment attempts to answer the following fundamental questions:

— what can happen and why (by risk identification)?

— what are the consequences?

— what is the probability of their future occurrence?

— are there any factors that mitigate the consequence of the risk or that reduce the probability of the risk?

— is the level of risk tolerable or acceptable and does it require further treatment?

This Standard is intended to reflect current good practices in selection and utilization of risk assessment techniques, and does not refer to new or evolving concepts which have not reached a satisfactory level of professional consensus.

This Standard is general in nature, so that it may give guidance across many industries and types of system. There may be more specific standards in existence within these industries that establish preferred methodologies and levels of assessment for particular applications. If these standards are in harmony with this Standard, the specific standards will generally be sufficient.

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